

Cloudy with a Chance of Machine Learning:

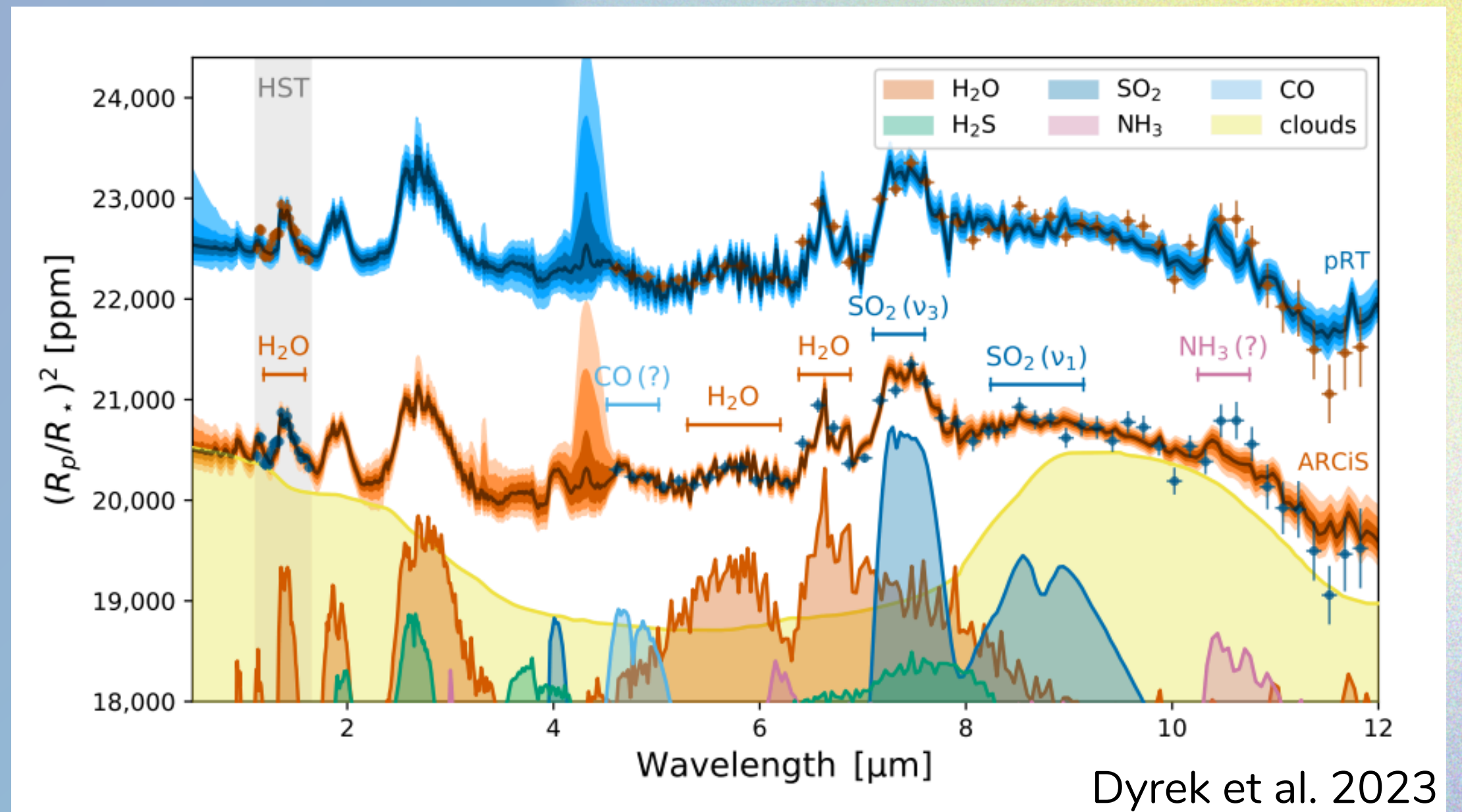
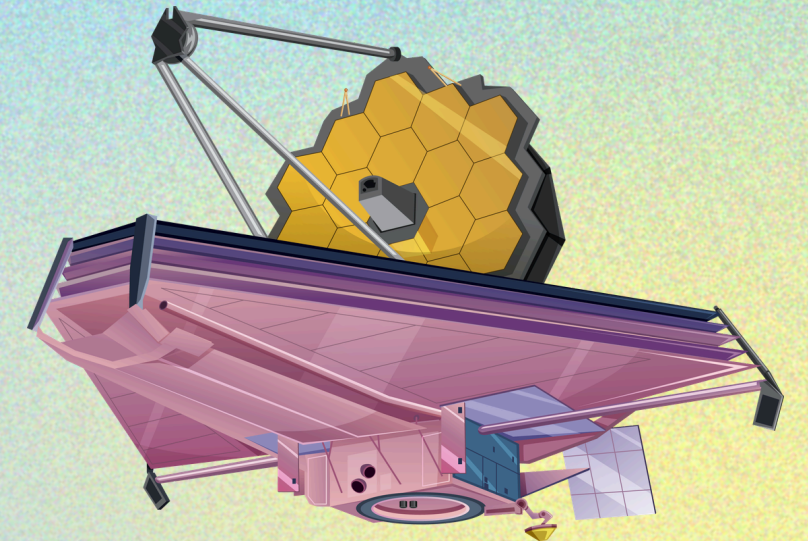
*Neural Network Predictions for
Well-Mixed Exoplanet Cloud
Particles*

Daisy Attaway

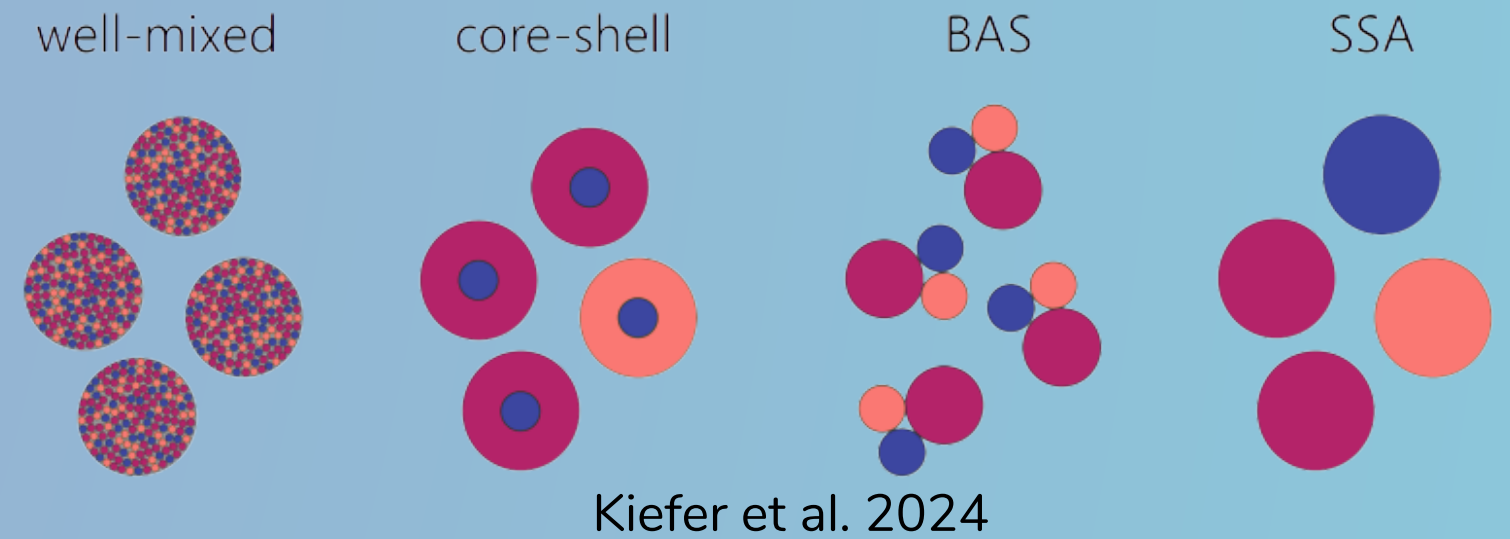
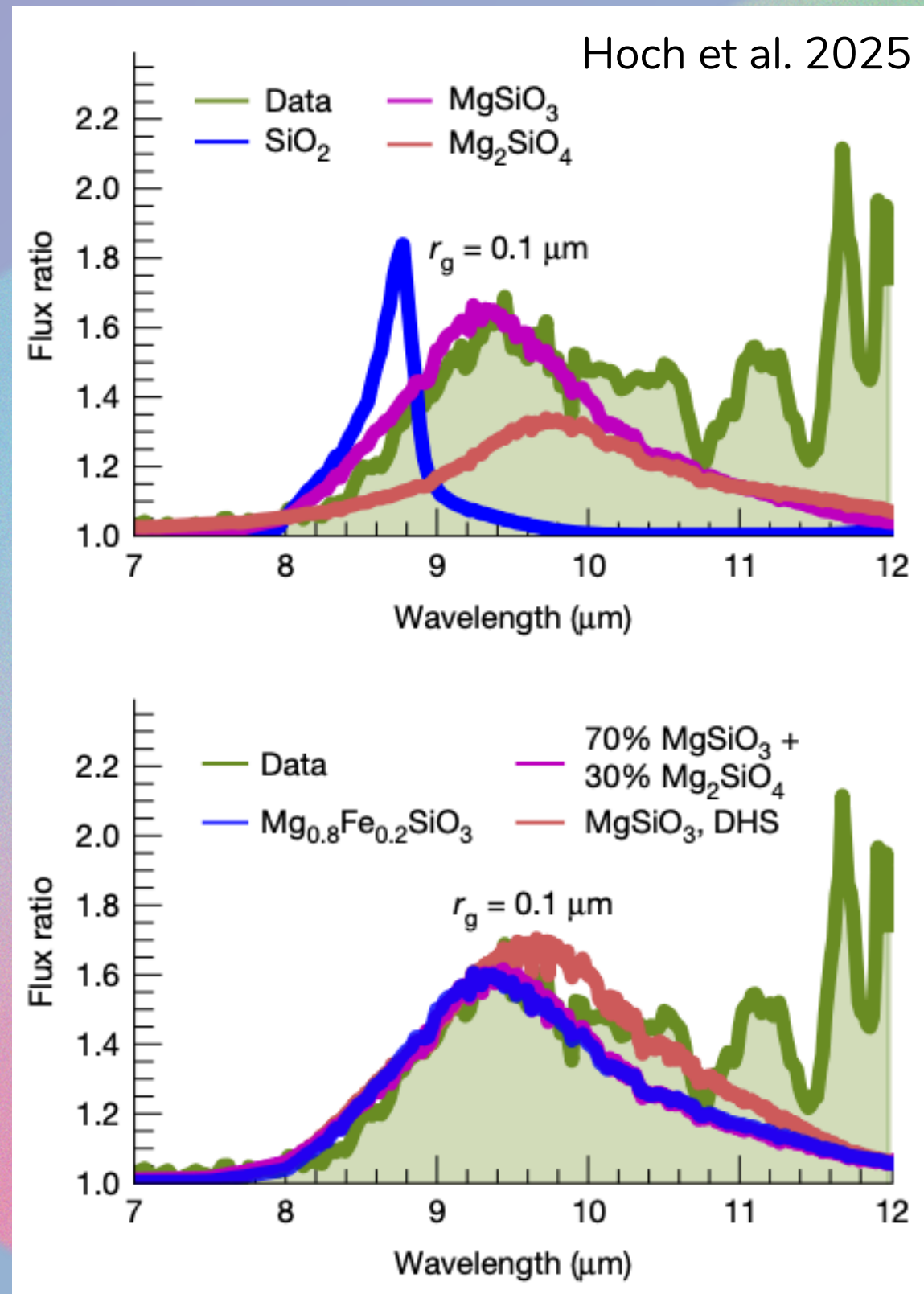
April 3, 2026

Motivation

- **Exoplanets**, or planets outside of our solar system, can have clouds that are very different than our clouds on Earth.
- With JWST it is now possible to observe a **diversity of cloud materials** in an Exoplanet's transmission spectra [1].



Motivation



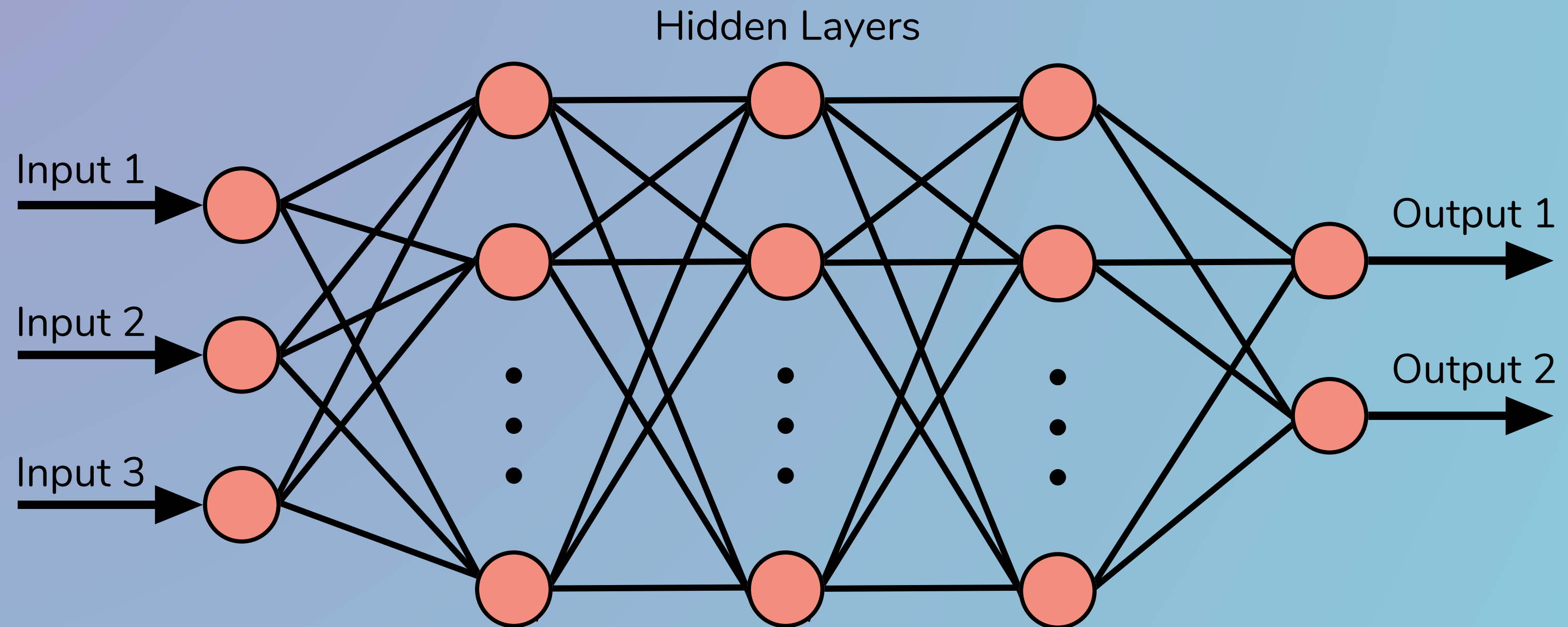
- Hoch et al 2025 finds including **mixtures of silicates species** resolves discrepancies between the spectral data and atmospheric models.
- There are a variety of mixing theories; we will focus on well-mixed cloud particles.
- Cloud **modeling is vital** to understanding Exoplanets and interpreting their observations; however, modeling mixed cloud particles is **computationally intensive**.

The Big Question...

*Can we use
machine learning
to optimize
modeling of mixed
cloud particles?*



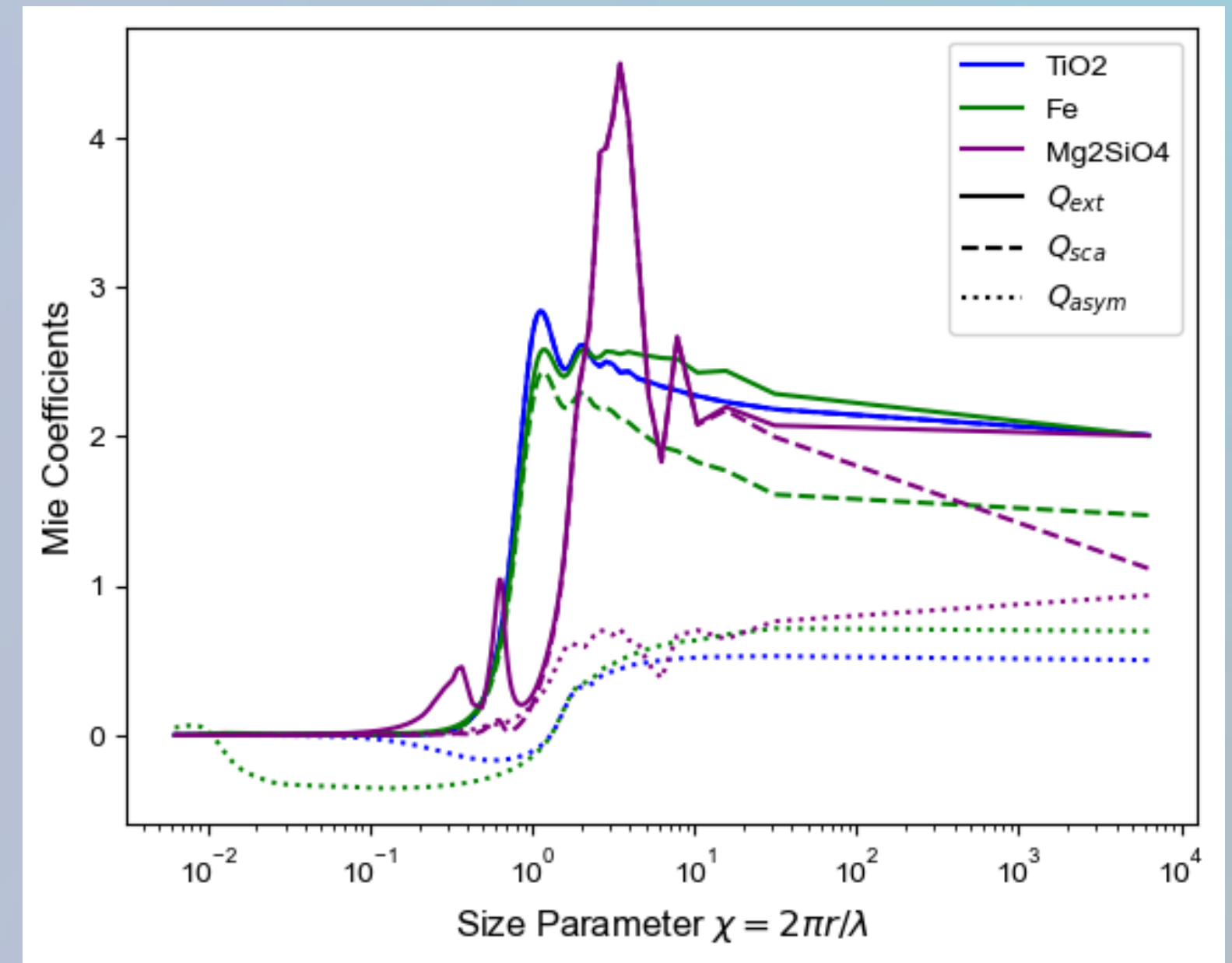
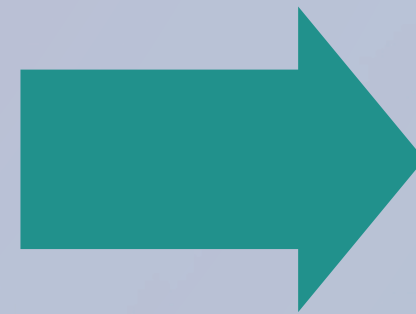
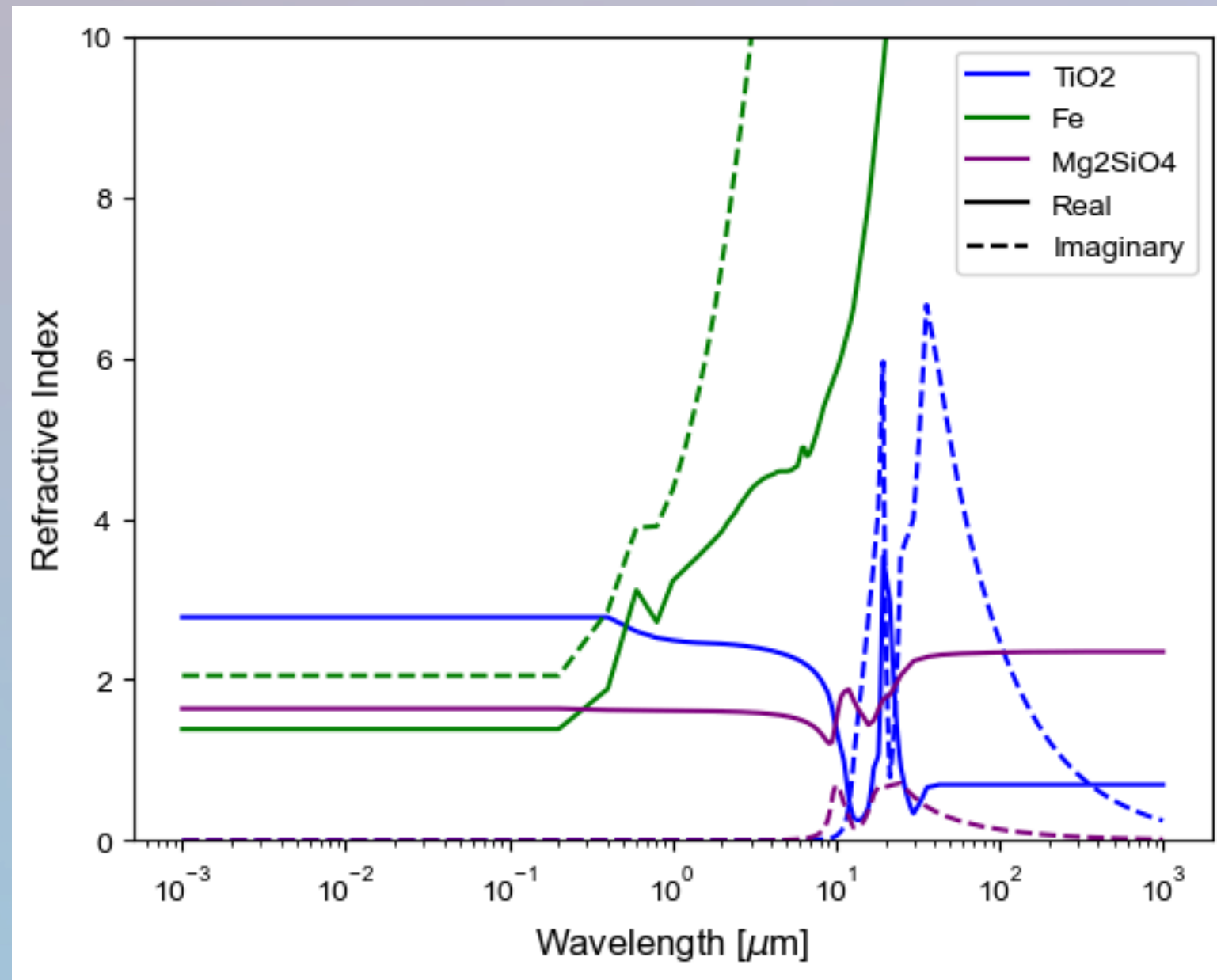
Neural Networks



- Uses neurons to process information, learn patterns in the data, and make predictions [2].
- Highly customizable and can handle **complex, non-linear data**.

Mie Theory

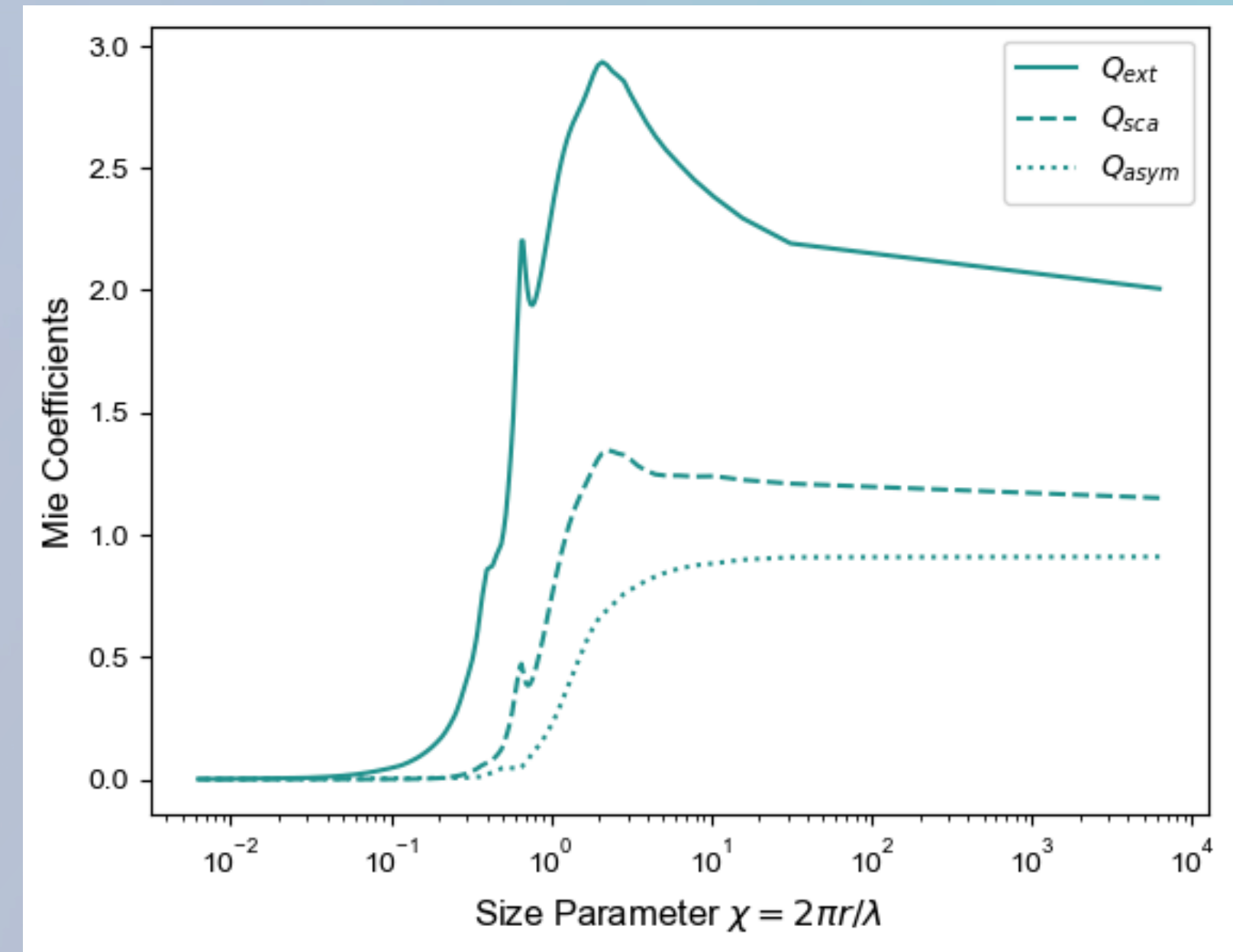
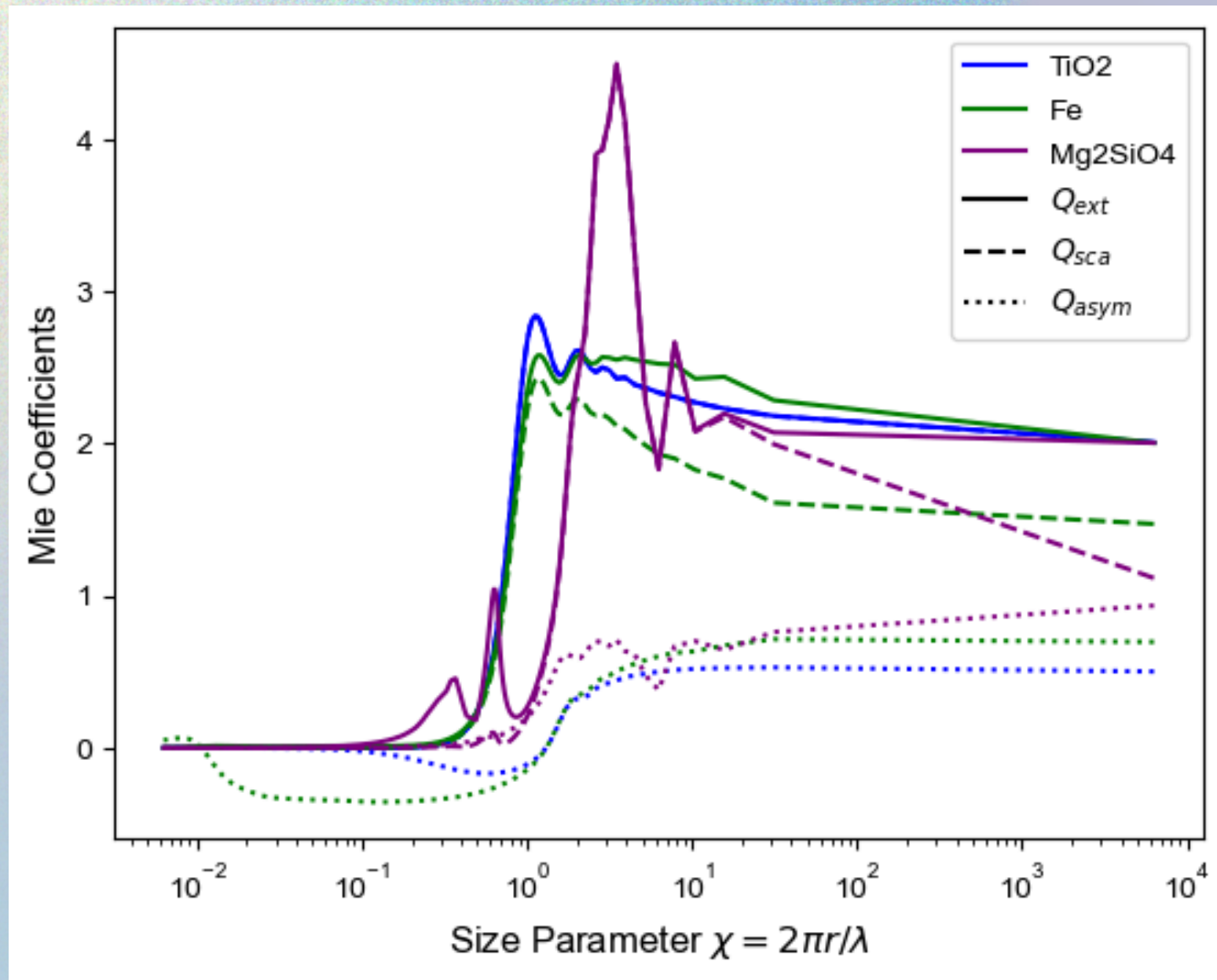
- Describes how light interacts with particles of a **similar size to the wavelength** of light [3].
- Results are described using the Extinction, Scattering, and Asymmetry parameters.



LLL Approximation

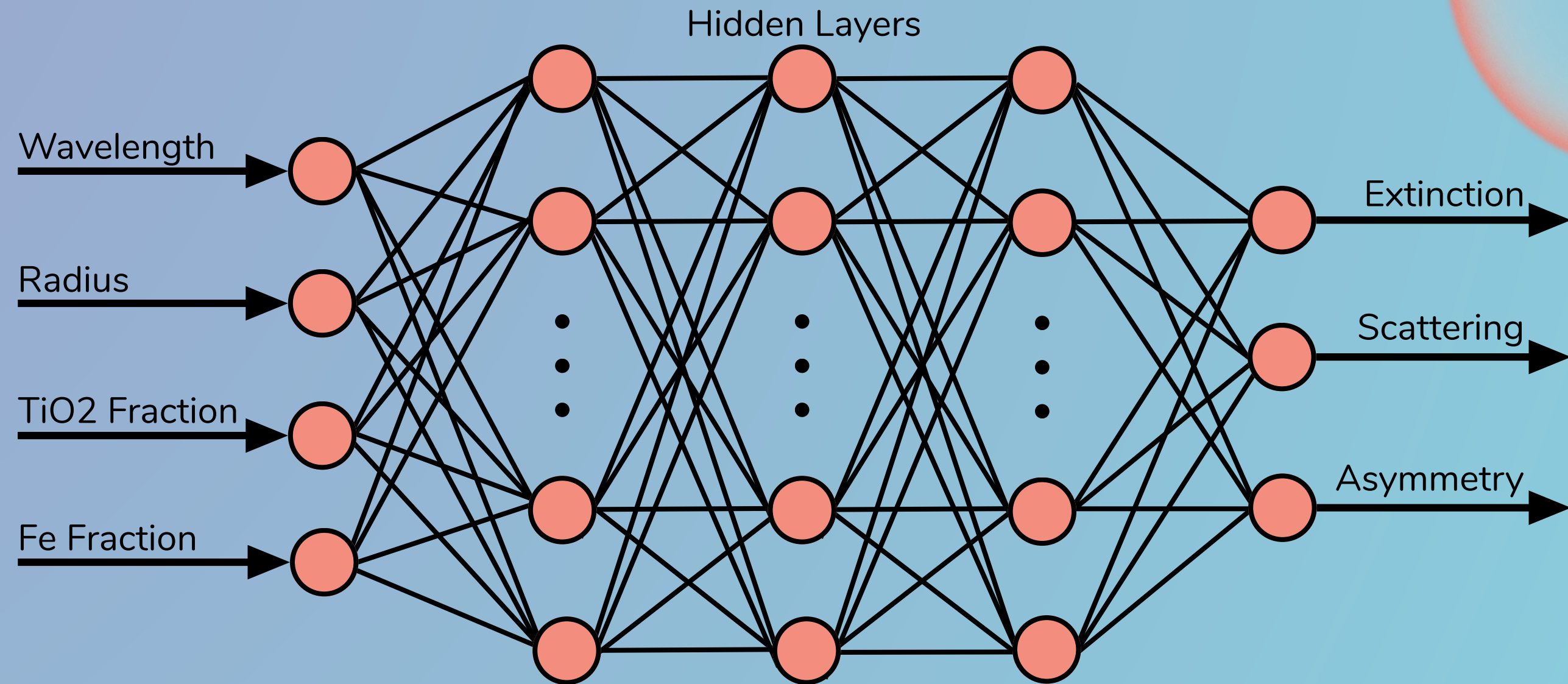
- Approximates an **effective refractive index** for a mixture of species [4].

$$\epsilon_{\text{eff}} = \left(\sum_i f_i \sqrt[3]{\epsilon_i} \right)^3$$



The Preliminary Model

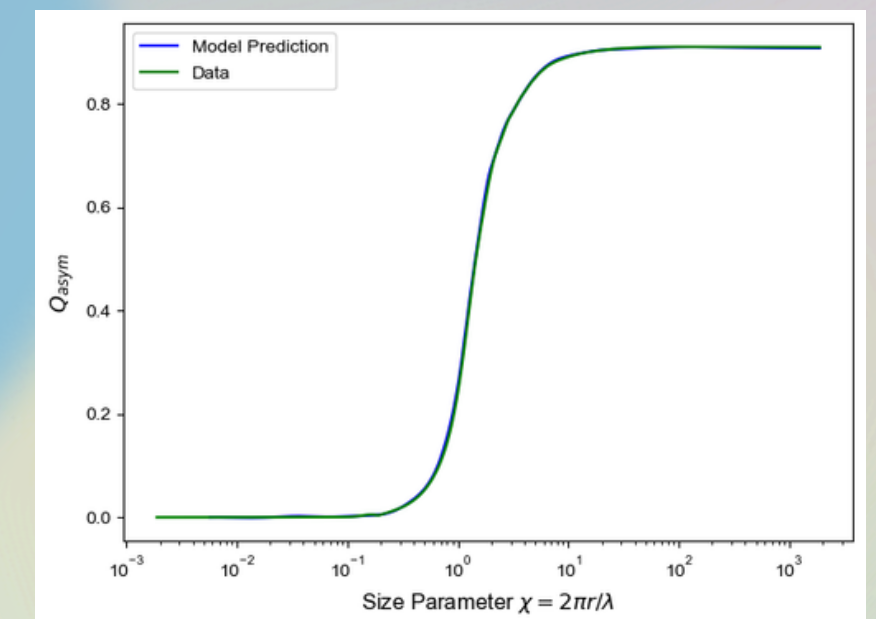
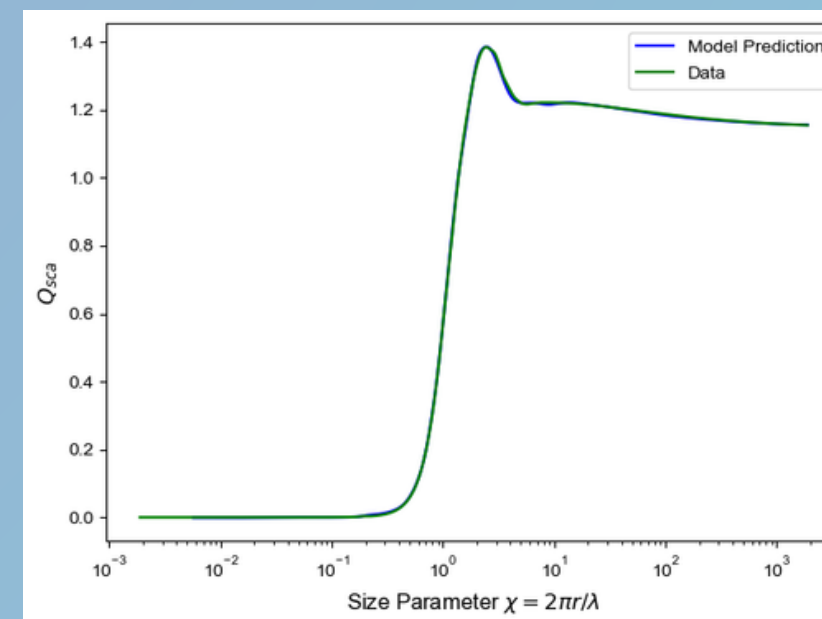
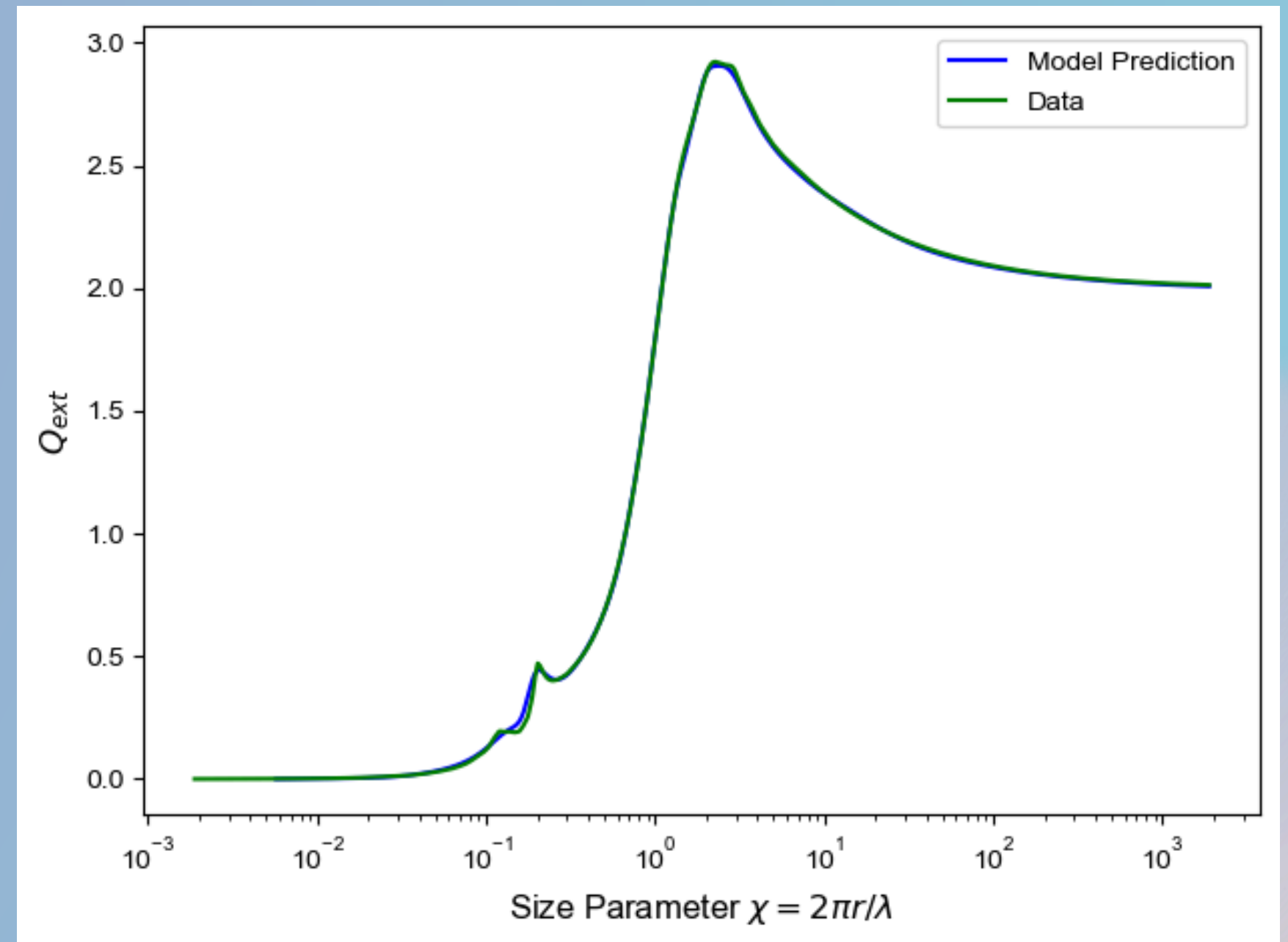
- To test the viability of using machine learning, we first train a model for a mixture of **TiO₂, Fe, and Mg₂SiO₄**.
- This mixture is physically realistic, computationally challenging, and should result in a **defined feature** in the model spectrum at a wavelength of 10 microns from Mg₂SiO₄.



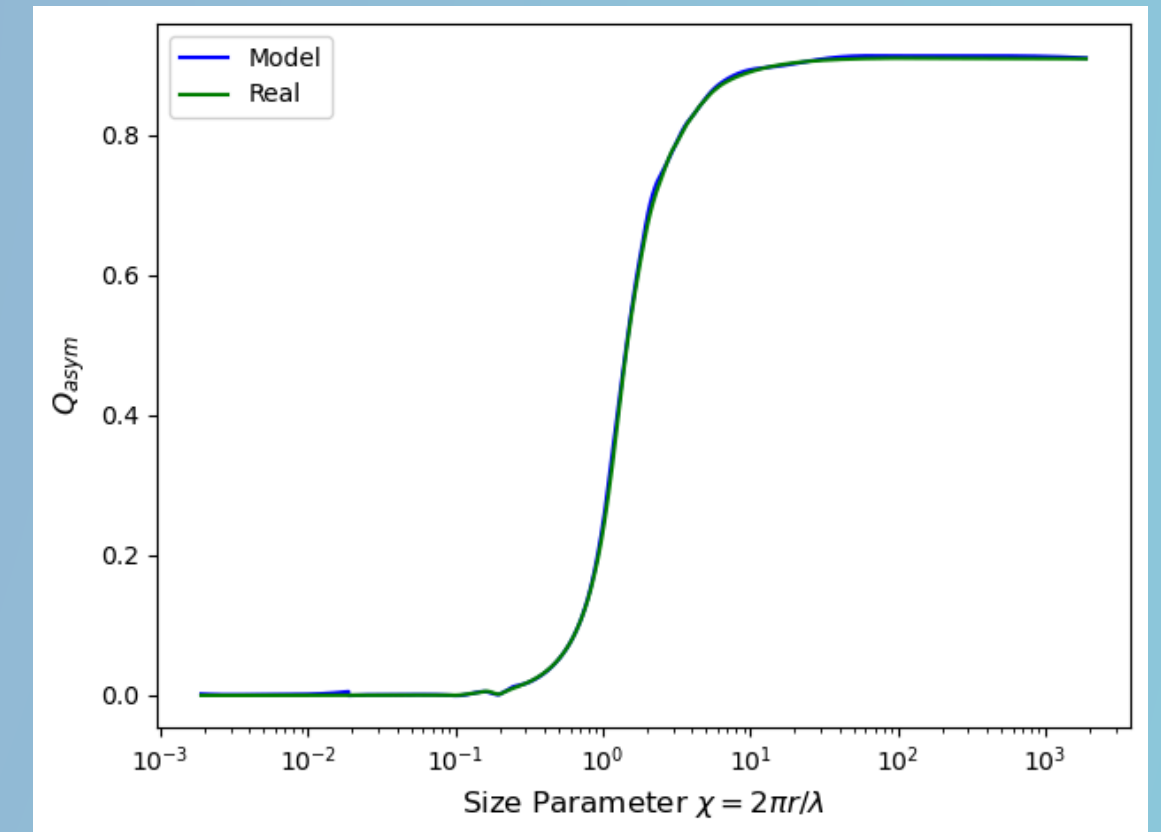
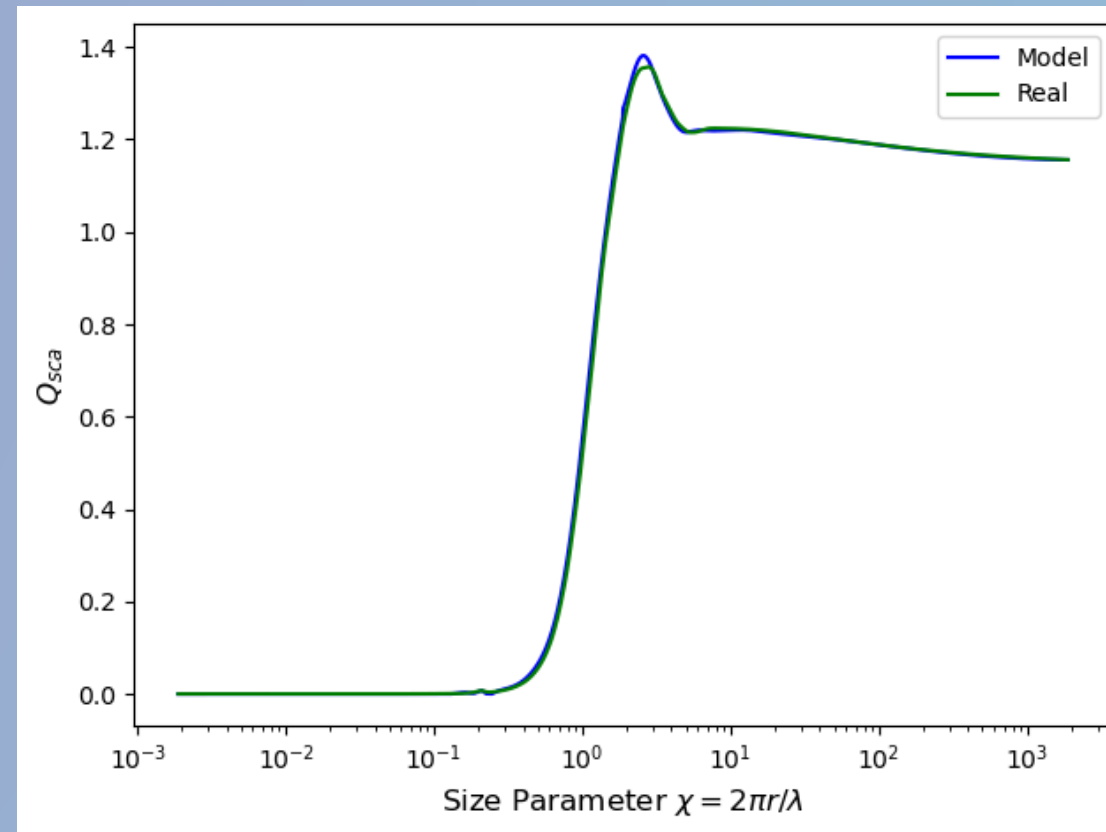
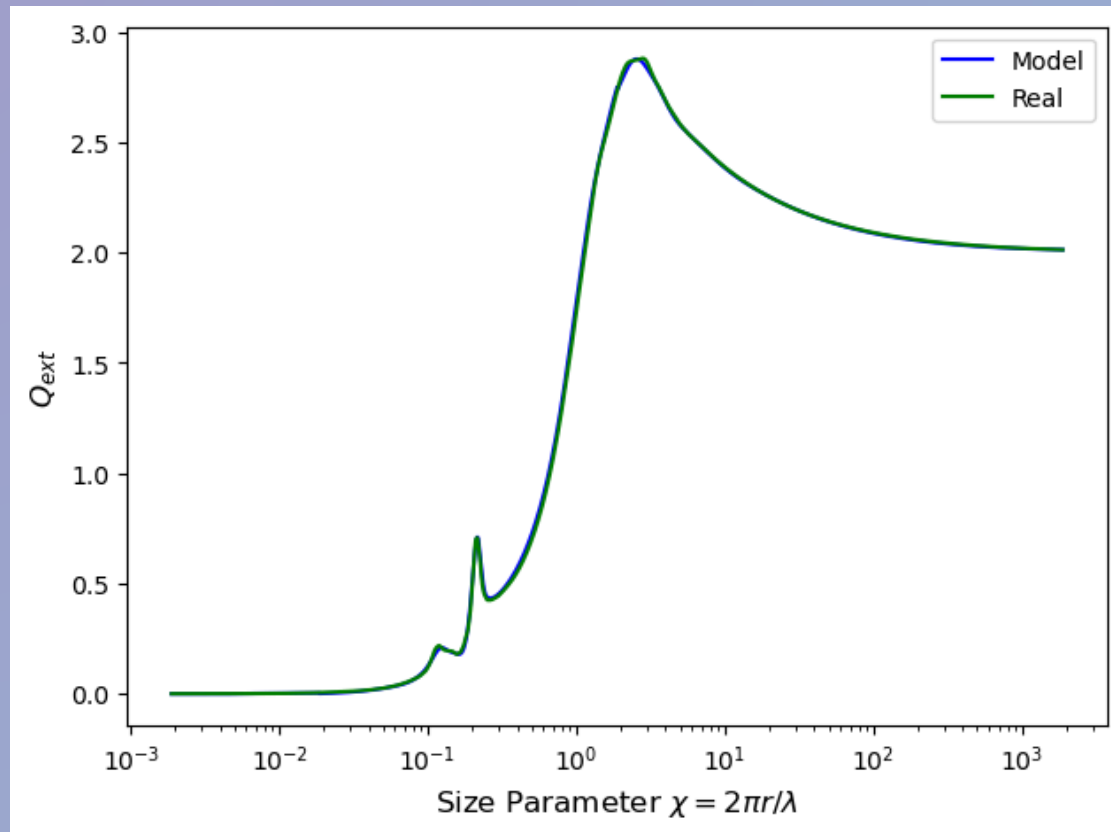
Results

- Prediction results for **Mg₂SiO₄-dominated** (90%) mixture with radius of 3 microns.
- We find that the network successfully fits the coefficient curves and **reproduces the 10 micron feature**.

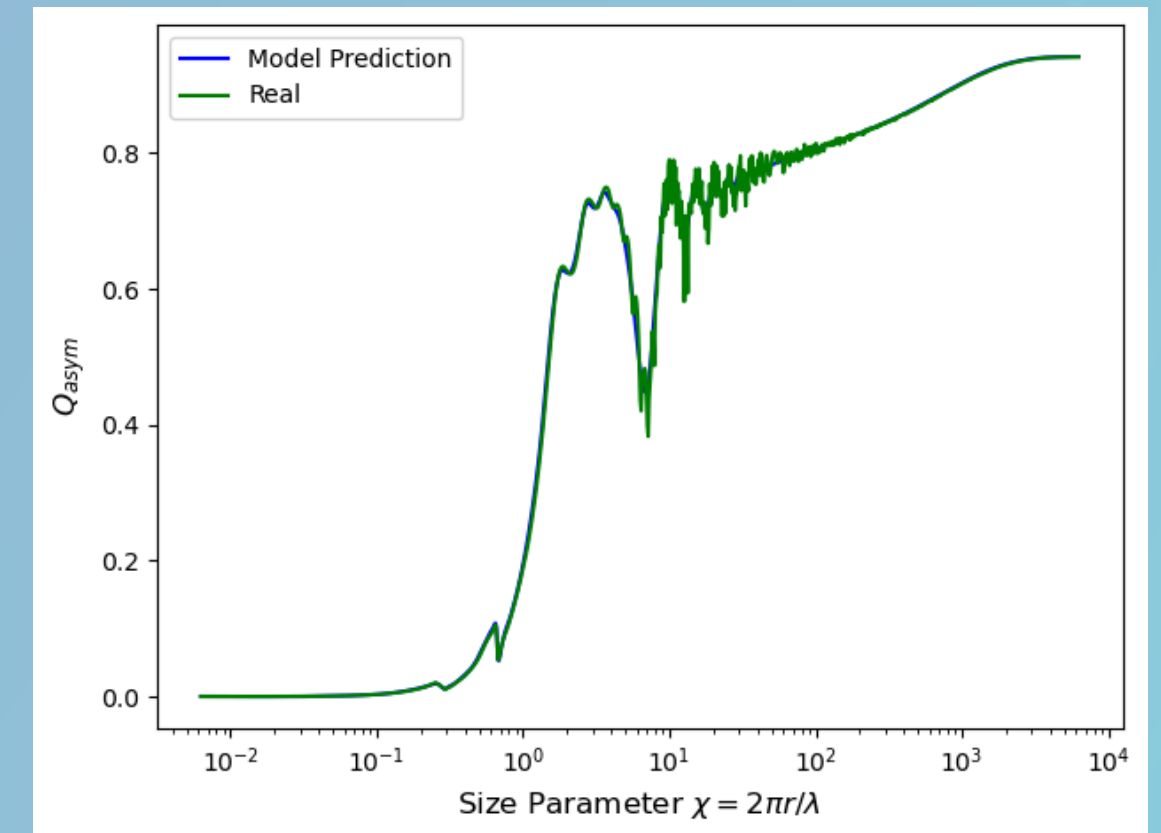
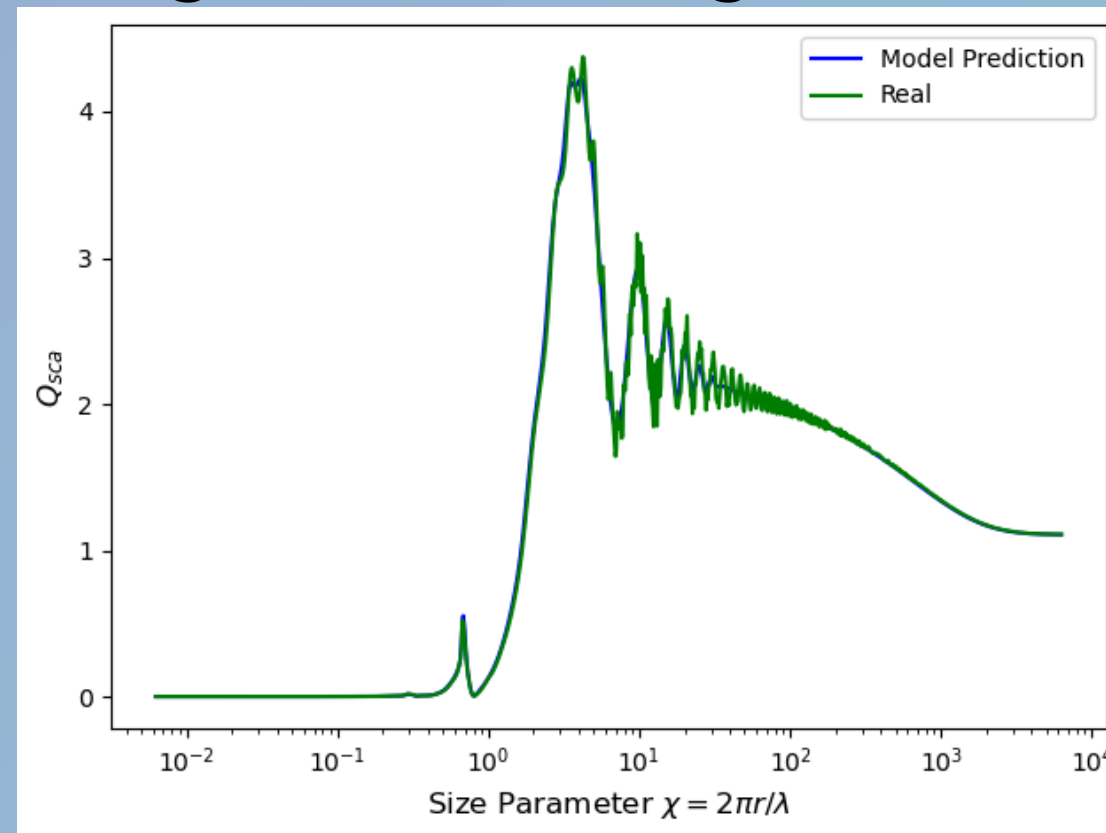
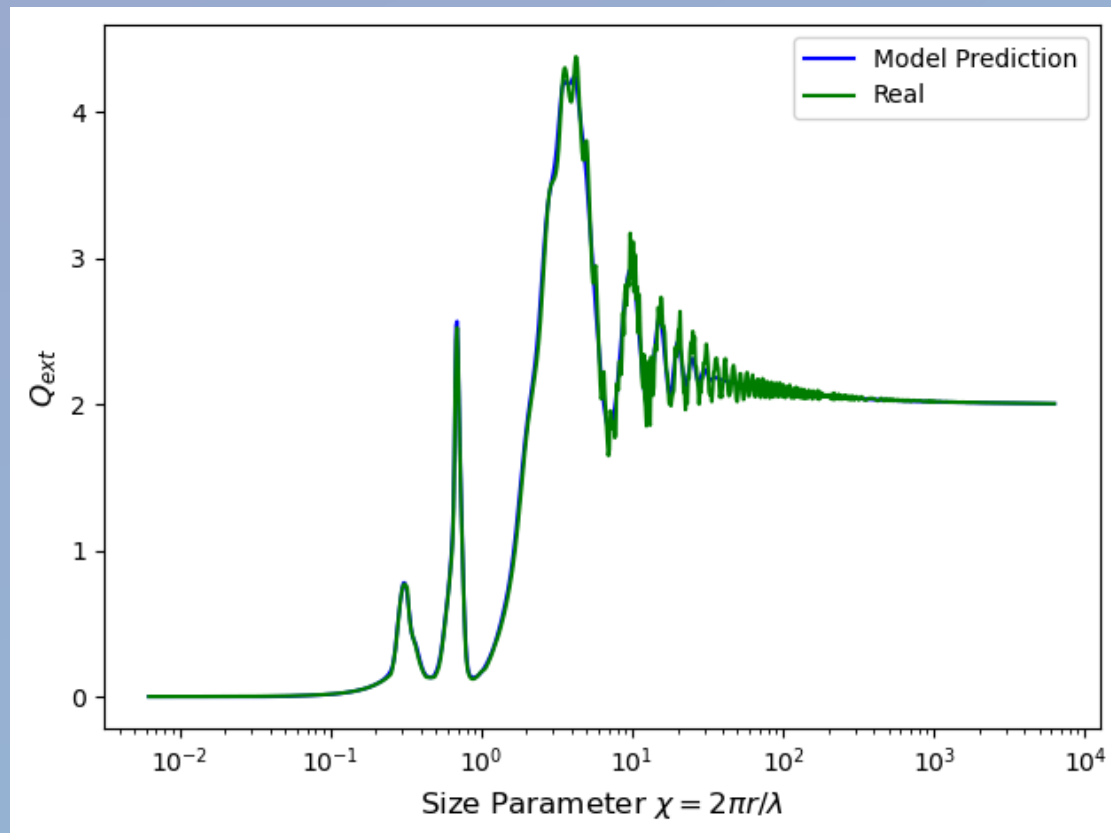
Computational time five times faster on average than normal calculations!



TiO₂, Fe, and MgSiO₃ Mixture (MgSiO₃ dominated)



SiO₂, MgSiO₃, and Mg₂SiO₄ Mixture



Take Home Message

- Transmission spectra from JWST observations show evidence of mixed cloud particles in exoplanet clouds.
- Atmospheric models are key to understanding exoplanets and their observations.
- Our neural network accurately reproduces mie coefficients and key optical features five times faster on average than normal calculations for 3 different mixtures of materials.

Future Directions

- Continue to extend this process to other mixtures.
- Make results of neural networks publicly available as a Python package.
- Integrate this work into the Virga cloud model.

Questions?

References:

- [1] Hoch, K.K.W., Rowland, M., Petrus, S. et al. Silicate clouds and a circumplanetary disk in the YSES-1 exoplanet system. *Nature* 643, 938–942 (2025).
- [2] LeCun, Y., Bengio, Y. & Hinton, G. Deep learning. *Nature* 521, 436–444 (2015).
- [3] Mie, G. (1908), Beiträge zur Optik trüber Medien, speziell kolloidaler Metallösungen. *Ann. Phys.*, 330: 377-445.
- [4] Landau, L. D., & Lifshitz, E. M. (1960). *Electrodynamics of continuous media*.